

School of Physics and Astronomy



Calculating Weyl anomaly on twistor space

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1 Review

In this review, I will describe the methods developed in [1] to calculate gauge anomalies and diffeomorphism anomalies on twistor space for self-dual Einstein manifolds and extend the work to calculate Weyl anomalies. In addition to these twistor space calculations, I have worked on renormalizing higher derivative classical gravity in the presence of quantum fields (discussed as a future project in my first-year review). I have also looked into the problem of doubling of chiral fermions in the context of Kahler-Dirac fermions and the spectral triple, but I refrain from discussing that in this review for the sake of

being brief.

2 Twistors

2.1 Introduction

Twistors are spinors of the complexified conformal group $Sl(4, \mathbb{C})$. They have a nice representation in terms of the spinors of the complexified 4D space $\mathcal{M}_{\mathbb{C}}$:

$$Z^\alpha = (\mu^{A'}, \lambda_A) \in \mathbb{PT}. \quad (1)$$

Here $\mu^{A'}$ and λ_A are left and right-handed spinors of $Spin(4, \mathbb{C})$ related to the 4D spacetime point by the incidence relation:

$$\mu^{A'} = x^{AA'} \lambda_A, \quad x^{AA'} \in \mathcal{M}_{\mathbb{C}}, \lambda \in \mathbb{CP}^1. \quad (2)$$

This provides an interesting perspective of looking at twistor space as the fibration of $\mu^{A'}$ over the celestial sphere coordinatized by λ_A with the sections of the fibre being the spacetime points.

Let's say $\mu^{A'} = (\mu^{0'}, \mu^{1'}) = (u, v)$ and $\lambda_A = (1, z) \in \mathbb{CP}^1$ then

$$Z^\alpha = (\mu^{A'}, \lambda_A) = (u, v, 1, z) \in \mathbb{CP}^3 \setminus \mathbb{CP}^1. \quad (3)$$

One can go to the anti-holomorphic patch where $Z^\alpha = (u/z, v/z, 1/z, 1) \approx (\bar{u}, \bar{v}, \bar{z}, 1)$ and therefore we infer that $\mu^{A'} = (u, v)$ lives in a $\mathcal{O}(1) \oplus \mathcal{O}(1)$ bundle over $\mathbb{CP}^1$. Therefore twistor space will be defined as a line bundle over the celestial sphere:

$$\mathbb{PT} \equiv \mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1. \quad (4)$$

As discussed before, complexified spacetime is the section of this bundle:

$$\mathcal{M}_{\mathbb{C}} \hookrightarrow \mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1. \quad (5)$$

Notice that the incidence relations are describing a 2 - plane $\mathbb{C}^2 \subset \mathbb{C}^4 \equiv \mathbb{T}$ in twistor space. Projectivizing these planes gives Riemann spheres $\mathbb{CP}^1$ (not to be confused with the base manifold in the fibration). Therefore, a spacetime point $x^{AA'}$ corresponds to a line L_x (a Riemann sphere) in twistor space.

What happens when two lines meet? Let's say we have $\mu^{A'} = x^{AA'} \lambda_A$ and $\mu^{A'} = y^{AA'} \lambda_A$ then subtracting the two equations gives $(x^{AA'} - y^{AA'}) \lambda_A = 0$, which means that $(x^{AA'} - y^{AA'})$ is proportional to λ_A and can be written as $(x^{AA'} - y^{AA'}) = \tilde{\lambda}^{A'} \lambda^A$. Further, note that $\det(\tilde{\lambda}^{A'} \lambda^A) = 0$, therefore the vector $(x^{AA'} - y^{AA'})$ must be a null-vector. λ_A is referred to as the flag pole of the null-vector, whereas $\tilde{\lambda}^{A'} \in \mathbb{C}^2$ is referred to as the flag plane or α - plane.

We conclude this subsection by highlighting that:

1. Spacetime points are lines (Riemann spheres) in twistor space

2. To lines intersect if the corresponding spacetime points are null-separated.
3. Points in twistor space (where any two lines meet) correspond to α – *planes* in spacetime.
4. Therefore, local spacetime points become global in twistor space, whereas local twistor points become global in spacetime.

2.2 Conformal structure and infinity twistor

Let X be a spacetime point and Z_1^α and Z_2^α be two points in twistor space that lie on L_X . Then the bitwistor $X^{\alpha\beta} := Z_1^{[\alpha} Z_2^{\beta]}$ is given by:

$$X^{\alpha\beta} = \langle \lambda_1 \lambda_2 \rangle \begin{bmatrix} \frac{1}{2} \epsilon^{A'B'} x^2 & x_B^{A'} \\ -x_A^{B'} & \epsilon_{AB} \end{bmatrix} \quad (6)$$

The bitwistor encodes all the information about the spacetime point up to a proportionality factor of $\langle \lambda_1 \lambda_2 \rangle = \epsilon_{AB} \lambda_1^A \lambda_2^B$.

For the sake of completeness, we would like to mention that the bitwistor lives in the space of Grassmannian 2 – *planes* of the twistor space $\Lambda^2(\mathbb{PT})$. The fact that points in this 6 – *dimensional* complex space $\mathbb{CP}^5$ are related to $\mathbb{CP}^1$ lines (that correspond to spacetime points) in twistor space $\mathbb{CP}^3$ is called the Klein-correspondance. All such special points in the $6D$ space lie on a special quadric called the Klein quadric and capture conformal compactification from 6 dimensions to 4 dimensions.

A natural $Sl(4, \mathbb{C})$ invariant inner-product on twistor space is:

$$\langle Z_1 Z_2 Z_3 Z_4 \rangle = \epsilon_{\alpha\beta\gamma\delta} Z_1^\alpha Z_2^\beta Z_3^\gamma Z_4^\delta. \quad (7)$$

Doing a short calculation, one can check that:

$$\epsilon_{\alpha\beta\gamma\delta} dX^{\alpha\beta} dX^{\gamma\delta} = \langle \lambda_1 \lambda_2 \rangle^2 dx^{AA'} dx_{AA'} = \langle \lambda_1 \lambda_2 \rangle^2 ds^2. \quad (8)$$

Thus the $Sl(4, \mathbb{C})$ invariant inner-product gives us the line element in spacetime up to a conformal factor of $\langle \lambda_1 \lambda_2 \rangle$. However, since $Sl(4, \mathbb{C})$ is the complexified conformal group, ds^2 is the same as $\langle \lambda_1 \lambda_2 \rangle^2 ds^2$.

To break conformal symmetry, we need to fix the asymptotics of our theory, which is achieved by choosing an infinity twistor. An infinity twistor is a bitwistor that fixes the asymptotics of the spacetime. We will modify the metric to be:

$$ds^2 = \frac{\epsilon_{\alpha\beta\delta\gamma} dX^{\alpha\beta} dX^{\gamma\delta}}{(I_{\alpha\beta} X^{\alpha\beta})^2} \quad (9)$$

To get asymptotically flat spacetimes we will choose the following infinity twistor:

$$I_{\alpha\beta} = \frac{1}{2} \begin{bmatrix} 0 & 0 \\ 0 & \epsilon^{AB} \end{bmatrix}, \quad (10)$$

then we have $I_{\alpha\beta} X^{\alpha\beta} = \langle \lambda_1 \lambda_2 \rangle$ and therefore $ds^2 = dx^{AA'} dx_{AA'}$. To choose, asymptotically de-Sitter or anti de-Sitter spacetimes one can choose the following infinity twistor:

$$I_{\alpha\beta} = \frac{1}{2} \begin{bmatrix} \Lambda \epsilon^{A'B'} & 0 \\ 0 & \epsilon^{AB} \end{bmatrix}. \quad (11)$$

2.3 Penrose-Ward transforms

Theorem(Penrose): Zero-rest mass chiral fields on spacetime correspond to homogenous and holomorphic (if on-shell) functions on twistor space.

We will provide a hand-wavy proof of the theorem below. The Penrose transform maps homogeneous functions on twistor space to chiral massless fields on spacetime. A negative helicity ($h = -1$), spin s field on spacetime is given by:

$$\phi_{A_1 \dots A_s}(x) = \int_{L_x \cong \mathbb{CP}_x^1} \langle \lambda d\lambda \rangle \lambda_{A_1} \lambda_{A_2} \dots \lambda_{A_s} f(Z) \Big|_{L_x} \quad (12)$$

where $\langle \lambda d\lambda \rangle \in \Omega^{(1,0)}(\mathbb{CP}^1)$ is the holomorphic top-form on $\mathbb{CP}^1$. Since the spacetime field $\phi_{A_1 \dots A_s}(x)$ is not homogeneous in the twistor variable λ_A , the term on the right-hand side should also not be homogeneous. Looking at the integrand, this is possible only if the function $f(Z)$ is of homogeneity $-2s - 2$. $f(Z)$ should also be the anti-holomorphic top-form for the integrated to be a $(1, 1)$ form integrated over $\mathbb{CP}^1$. We will denote these requirements as:

$$f(Z) \in \Omega^{(0,1)}(\mathbb{PT}, \mathcal{O}(-2s - 2)). \quad (13)$$

The equation of motion for the negative helicity, spin s field on spacetime is given by:

$$\partial_{B'}^{A_1} \phi_{A_1 \dots A_s}(x) = 0. \quad (14)$$

On the twistor side, we use the incidence relation to write this equation as:

$$\int_{L_x \cong \mathbb{CP}_x^1} \langle \lambda d\lambda \rangle \lambda_{A_1} \lambda_{A_2} \dots \lambda_{A_s} \left[\lambda^{A_1} \frac{\partial f(Z)}{\partial \mu^{B'}} + \bar{\lambda}^{A_1} \frac{\partial f(Z)}{\partial \bar{\mu}^{B'}} \right] \Big|_{L_x} = 0. \quad (15)$$

The first term inside the bracket vanishes because $\lambda_{A_1} \lambda^{A_1}$ is zero. Therefore, the second term must be zero for the equation of motion to hold true.

$$\bar{\lambda}^{A_1} \frac{\partial f(Z)}{\partial \bar{\mu}^{B'}} = 0 \quad (16)$$

which is the requirement that the function $f(Z)$ should be holomorphic, $\bar{\partial}f = 0$. It can be shown that if the function $f(Z)$ is exact in the Dolbeault operator, i.e., $f(Z) = \bar{\partial}g$ then the spacetime field is exactly zero, $\phi_{A_1 \dots A_s} = 0$. Therefore, the function $f(Z)$ should be closed but not exact, and we say that it lives in the Dolbeault cohomology:

$$f(Z) \in H^{(0,1)}(\mathbb{PT}, \mathcal{O}(-2s - 2)). \quad (17)$$

It is easy to extend this to fields with positive helicity and scalars, in which case, the Penrose transform looks like:

$$h = 0 \quad \phi(x) = \int_{L_x} \langle \lambda d\lambda \rangle f(Z) \Big|_{L_x} \quad (18)$$

$$h = 1 \quad \phi_{A'_1 A'_2 \dots A'_s}(x) = \int_{L_x} \langle \lambda d\lambda \rangle \frac{\partial}{\partial \mu^{A'_1}} \frac{\partial}{\partial \mu^{A'_2}} \dots \frac{\partial}{\partial \mu^{A'_s}} f(Z) \Big|_{L_x} \quad (19)$$

2.3.1 Gauge theories

Theorem(Ward [2]): A holomorphic $rank - N$ vector bundle on twistor space corresponds to a self-dual $Gl(N, \mathbb{C})$ Yang-Mills instanton moduli space on spacetime.

Before we discuss this theorem, let us review some basic concepts related to complex vector spaces. A complex vector space V is equipped with an almost complex structure J_b^a , a map from the vector space to itself. J should square to the identity map, i.e., $J_b^a J_c^b = \delta_c^a$. One can define a curvature associated with the almost complex structure called the Nijenhuis tensor, which is given by:

$$N_{bc}^a = J_b^d \partial_d J_c^a - J_c^d \partial_d J_b^a + J_d^a (\partial_b J_c^d - \partial_c J_b^d). \quad (20)$$

If the Nijenhuis tensor vanishes, we say that the almost complex structure is integrable (and hence a complex structure). An almost complex structure is needed to define the (anti-)holomorphic derivative $\bar{\partial}$. Vanishing of the Nijenhuis tensor is encapsulated in the nilpotency of the (anti-)holomorphic derivative operator $\bar{\partial}^2 = 0$.

To define gauge theories on twistor space, we need to $rank - N$ principle bundle on twistor space $E : \mathbb{C}^N \rightarrow \mathbb{PT}$. The connection on this bundle will be defined using the endomorphisms of the fibre, $End(\mathbb{C}^N) \cong Gl(N, \mathbb{C})$. We define the following connection:

$$\bar{D} = \bar{\partial} + A, \quad (21)$$

where $A \in \Omega^{(0,1)}(\mathbb{PT}, Gl(N, \mathbb{C}))$.

The connection $\bar{D}$ describes an almost complex structure on twistor space. The almost complex structure will be complex when the curvature tensor defined using this connection vanishes, i.e.,

$$F^{(0,2)} = [\bar{D}, \bar{D}] = \bar{\partial}A + \frac{1}{2}[A, A] = 0. \quad (22)$$

Ward's theorem precisely relates the vanishing curvature tensor to solutions of self-dual Yang-Mills instantons.

Recall that the Yang-Mills curvature can be decomposed into self-dual and anti-self-dual bits.

$$F_{AA'BB'} = \epsilon_{A'B'} F_{AB} + \epsilon_{AB} F_{A'B'} \quad (23)$$

Self-dual Yang-Mills is given by $F_{A'B'} = 0$ or $F_{AA'BB'} = \epsilon_{A'B'} F_{AB}$.

2.3.2 The non-linear graviton

Theorem(Penrose [3]): There is a correspondence between curved manifolds with a conformal class of metrics $[g]$ and self-dual Weyl tensor and complex 3 - folds $\mathcal{PT}$ that are complex deformations of the twistor space $\mathbb{PT}$ with at least one rational curve L_x (basically a $\mathbb{CP}^1$) with normal bundles $\mathcal{O}(1) \oplus \mathcal{O}(1)$.

The deformation of the flat twistor space is captured by the almost complex structure:

$$\bar{\partial} \rightarrow \bar{\nabla} = \bar{\partial} + \mathcal{L}_V \quad (24)$$

where V is a $(0, 1)$ -form valued in the tangent space $T_{\mathcal{PT}}$ of the twistor space.

$$V = V_{\bar{\beta}}^{\alpha} d\bar{Z}^{\beta} \frac{\partial}{\partial Z^{\alpha}} \in \Omega^{(0,1)}(\mathcal{PT}, T_{\mathcal{PT}}). \quad (25)$$

The field strength tensor for the connection $\bar{\nabla}$ is given by:

$$N^{\alpha} = \bar{\partial}V^{\alpha} + \frac{1}{2}V^{\beta} \wedge \partial_{\beta}V^{\alpha} \in \Omega^{(0,2)}(\mathcal{PT}) \quad (26)$$

$$N = N^{\alpha} \frac{\partial}{\partial Z^{\alpha}} \in \Omega^{(0,2)}(\mathcal{PT}, T_{\mathcal{PT}}) \quad (27)$$

where $V^{\alpha} = V_{\bar{\beta}}^{\alpha} d\bar{Z}^{\beta} \in \Omega^{(0,1)}(\mathcal{PT})$.

The vanishing of this field-strength tensor ensures that the Nijenhuis tensor for the almost complex structure vanishes, making it integrable.

We state without proof that the vanishing of the Nijenhuis tensor on twistor space corresponds to the vanishing of the anti-self-dual Weyl tensor on spacetime. The problem that we obtain only self-dual Weyl tensor on spacetime from twistor construction is often referred to as the Googly problem.

3 BF action

3.1 Self-dual Yang-Mills

We will now try to construct actions for self-dual Yang-Mills and self-dual Weyl gravity on twistor space. First, let us consider the action for self-dual Yang-Mills. Note that twistor space is a complex 3-space and therefore, the action must be a $(3, 3)$ form integrated over twistor space. Consider the following action:

$$S = \int_{\mathbb{PT}} b \wedge F \quad (28)$$

where $F = \bar{\partial}a + \frac{1}{2}[a, a] \in \Omega^{(0,2)}(\mathbb{PT}, \mathfrak{g})$ is the field strength of the connection $\bar{D} = \bar{\partial} + a$. For the integral to be a top-form on twistor space, we must have $b \in \Omega^{(3,1)}(\mathbb{PT}, \mathfrak{g}^*)$ for a background field b . We define $\mathfrak{g}^*$ as the dual of the Lie algebra such that the action is essentially traced over the gauge indices.

Variation of the action with the fields “ b ” and “ a ” give the following equations of motion:

$$F = 0 \quad (29)$$

$$Db = 0. \quad (30)$$

Using Ward’s theorem, we note that the equation of motion $F = 0$ corresponds to a self-dual Yang-Mills instanton in spacetime.

Let us interpret the other equation of motion $Db = 0$. First note that $b \in \Omega^{(3,1)}(\mathbb{PT}, \mathfrak{g}^*) \cong \Omega^{(0,1)}(\mathbb{PT}, \mathbb{K} \otimes \mathfrak{g}^*)$, where $\mathbb{K} \cong \Omega^{(3,0)}$ is the holomorphic top-form on twistor space. The holomorphic top-form is just a line bundle $\mathcal{O}(-4)$ on twistor space. To see this, let us work

in the homogeneous coordinates $Z^\alpha = (u, v, z, 1) \approx (u/z, v/z, 1, 1/z) = (\bar{u}, \bar{v}, 1, \bar{z}) \in \mathbb{CP}^3$. The holomorphic top-form is given by $d\Omega = du \wedge dv \wedge dz$ in the holomorphic patch, however, $d\bar{u} = du/z$, $d\bar{v} = dv/z$, and $d\bar{z} = -dz/z^2$, therefore $d\bar{\Omega} = d\Omega/z^4$.

Hence we conclude that $b \in \Omega^{(0,1)}(\mathbb{PT}, \mathcal{O}(-4) \otimes \mathfrak{g}^*)$. It is quite remarkable that $-4 = -2s - 2$ for $s = 1$ and therefore the b field corresponds to a negative helicity, spin 1 field by the Penrose transform. Also note that $a \in \Omega^{(0,1)}(\mathbb{PT}, \mathcal{O}(0) \otimes \mathfrak{g})$ corresponds to a spin 1 field with positive helicity, since $2s - 2 = 0$ for $s = 1$.

We conclude that the equations of motion correspond to a self-dual Yang-Mills instanton with a propagating anti-self-dual background field “ b ”.

The action described above is invariant under the following standard gauge transformations:

$$\begin{aligned} a &\rightarrow g\bar{D}g^{-1}, & F &\rightarrow gFg^{-1}, & b &\rightarrow gb g^{-1} \\ \delta a &= \bar{D}c, & \delta F &= [F, c], & \delta b &= [b, c] \end{aligned} \quad (31)$$

For some gauge parameter $c \in \Omega^{(0,0)}(\mathbb{PT})$. Note, however, that the BF action is also invariant under the following additional gauge transformation that follows from Bianchi identity:

$$\delta F = 0, \quad \delta b = \bar{D}d \quad (33)$$

for some $d \in \Omega^{(3,0)}(\mathbb{PT})$. In total, we have the following infinitesimal gauge transformations:

$$\delta a = \bar{D}c, \quad \delta d = \bar{D}d + [b, c]. \quad (34)$$

The BF action that we have described is gauge-invariant, but it is not gauge-fixed. To gauge-fix the action we will use the BRST symmetry, which is a supersymmetry of the action obtained by Grassmannification of the gauge parameter. Note that due to the gauge invariance, the action is already BRST invariant $\delta_B S = 0$, or in other words, closed under the BRST operator δ_B . We have the freedom to add BRST exact terms $\delta_B \Psi$ to the action, since these terms are trivially closed due to the nilpotency of the BRST operator $\delta_B^2 = 0$. There are multiple ways of adding the BRST exact term to fix the gauge. We provide some examples below.

The Faddeev-Popov gauge fixing procedure requires introducing additional auxiliary fields, the anti-ghost field $\bar{c} \in \Omega^{(3,3)}$ (not the anti-particle of the ghost field c that used to be the gauge parameter), and the Nakanishi-Lautrup field $m \in \Omega^{(3,3)}$. These auxiliary fields are introduced such that $\delta_B \bar{c} = m$, and $\delta_B m = \delta_B^2 \bar{c} = 0$. Henceforth, we use the notation, $\bar{\partial}^\dagger = *\bar{\partial}*: \Omega^{(p,q)} \rightarrow \Omega^{(p,q-1)}$ to denote the co-exterior derivative. Consider the following exact term:

$$\Psi = \bar{c}(\bar{\partial}^\dagger a - \xi * m) \quad (35)$$

$$\delta_B \Psi = \delta_B \bar{c}(\bar{\partial}^\dagger a - \xi * m) + \bar{c} \delta_B(\bar{\partial}^\dagger a) \quad (36)$$

$$= m(\bar{\partial}^\dagger a - \xi * m) + \bar{c} \bar{\partial}^\dagger \bar{D}c. \quad (37)$$

One can add this BRST exact term to the action, integrating out the Nakanishi-Lautrup field m gives us the term $(\bar{\partial}^\dagger a) * (\bar{\partial}^\dagger a)/2\xi$ in the action which is a Gaussian distribution

in the path-integral centered around the Lorentz gauge with the width of the distribution given by ξ .

Another interesting choice is:

$$\Psi = \bar{c}\bar{\partial}^\dagger a \quad (38)$$

$$\delta_B \Psi = m\bar{\partial}^\dagger a + \bar{c}\bar{\partial}^\dagger \bar{D}c. \quad (39)$$

In this choice, integrating out the Nakanishi-Lautrup field directly gives Lorentz gauge $\bar{\partial}^\dagger a = 0$. This choice has another benefit: It makes the gauge-fixed action look like a BF action with the b -field and the F -field being polyform fields [4].

3.2 Self-dual Weyl gravity

Just like Yang-Mills, the action for self-dual Weyl gravity is also a BF action. It is given by:

$$S = \int_{\mathbb{PT}} G_\alpha \wedge (\bar{\partial}V^\alpha + \frac{1}{2}V^\beta \wedge \partial_\beta V^\alpha) \quad (40)$$

where just as before $N^\alpha = \bar{\partial}V^\alpha + \frac{1}{2}V^\beta \wedge \partial_\beta V^\alpha \in \Omega^{(0,2)}(\mathcal{PT}, T_{\mathcal{PT}})$ and $G_\alpha \in \Omega^{(3,1)}(\mathcal{PT}, T_{\mathcal{PT}}^*)$. The dynamical fields are V^α and G_α , variation of the action with respect to the background field G_α simply gives $N^\alpha = 0$ which by the non-linear graviton construction is related to the vanishing of the anti-self-dual Weyl tensor.

The equation of motion obtained by the variation of V^α is given by $\bar{\partial}G_\alpha + V^\beta \wedge \partial_\beta G_\alpha = 0$. This essentially corresponds to an anti-self-dual background Weyl tensor propagating in spacetime. Note that $G_\alpha \in \Omega^{(3,1)}(\mathcal{PT}, T_{\mathcal{PT}}^*) \cong \Omega^{(0,1)}(\mathcal{PT}, \mathbb{K} \otimes T_{\mathcal{PT}}^*)$, where $\mathbb{K}$ is the holomorphic top-form or equivalently an $\mathcal{O}(-4)$ line bundle. The relation of the fields V^α and G_α with positive and negative helicity gravitons via the Penrose transform will be clearer when we review the self-dual Einstein gravity action.

It was proved by Mason[5] that the action given above is classically equivalent to the action of self-dual Weyl gravity. Analogous to the BF action for Yang-Mills, the action defined above is invariant under the following gauge transformations:

$$\delta V^\alpha = \bar{\partial}f^\alpha + V^\beta \wedge \partial_\beta f^\alpha := \bar{D}f^\alpha, \quad \delta G_\alpha = V^\beta \wedge \partial_\beta f^\alpha + \bar{D}\chi^\alpha. \quad (41)$$

To gauge fix this action, we must add BRST exact terms just like in the Yang-Mills case. So far, we have described the Yang-Mills and self-dual Weyl gravity action; both of these actions are classically invariant under BRST supersymmetry corresponding to the gauge group and Weyl transformations. However, the quantum effective action, which is obtained by integrating out these fields, is not invariant under these BRST transformations. The variation of the effective action under the BRST transformation is called the gauge anomaly. In the next section, we will calculate the Weyl anomaly contribution for fields of different spin in twistor space.

Note that instead of working with self-dual Weyl gravity which is Weyl invariant and follows more naturally on twistor space, we could also have worked with self-dual Einstein gravity. Self-dual Einstein gravity is no longer Weyl invariant and follows from fixing the asymptotics of the theory (essentially fixing an infinity twistor). SD Einstein gravity is still diffeomorphism invariant, and we can calculate the anomaly to diffeomorphisms in this theory; this has been done in [1], and we will review this later.

4 Weyl anomaly on twistor space

4.1 Atiyah-Singer index theorem (weaker)

Theorem[Atiyah-Singer] The index of the Dirac operator in in $2m$ dimensional Euclidean space is given by the m -th Chern character of its curvature.

$$ind(i\mathcal{D}) := \nu_+ - \nu_- = Ch_m(F) := \int_{\mathcal{M}} \frac{1}{m!} Tr[F \wedge \dots \wedge F]. \quad (42)$$

The physics proof requires using the Fujikawa method to compute the chiral anomaly by variation of the fermionic path integral measure[6]. The chiral anomaly in $2m$ dimensions is given by the m -th Chern character or equivalently the index of the Dirac operator. A stronger version of this theorem for arbitrarily curved Riemannian spaces is discussed later.

4.2 Gauge anomalies

In this section, we will see how gauge anomalies in $2m$ dimensions are related to chiral anomalies in $2m + 2$ dimensions through the method of descent. First, note that the quantum effective action is given by, and its BRST variation is the gauge anomaly:

$$e^{-W} = \int [d\psi d\bar{\psi} dA] e^{-S[A, \psi, \bar{\psi}]} \quad (43)$$

$$\delta_B W = \mathcal{A}; \quad \delta_B S = 0. \quad (44)$$

By construction, the gauge anomaly is BRST exact and therefore trivially close:

$$\delta_B \mathcal{A} = 0. \quad (45)$$

This is known as the Wess-Zumino consistency condition. The expression for gauge anomaly must be i) topological (does not depend on metric), ii) BRST exact.

Consider the Chern-Simons form Q_{2m+1} in $2m + 2$ dimensions such that $dQ_{2m+1}(A, F) = Ch_m(F)$. The Chern character is an invariant class and therefore invariant under BRST transformations:

$$\delta_B Ch_m(F) = 0 \quad (46)$$

$$\Rightarrow \delta_B dQ_{2m+1} = 0 \quad (47)$$

$$\Rightarrow -d\delta_B Q_{2m+1} = 0 \quad (48)$$

$$\Rightarrow \delta_B Q_{2m+1} = dP_{2m}^1(c, A, F) \quad \text{locally} \quad (49)$$

Where $P_{2m}^1(c, A, F)$ is a $2m$ -form that is topological and BRST exact, the superscript 1 denotes that it linearly depends on the ghost field c . P_{2m}^1 is the perfect candidate for the gauge anomaly. This way of obtaining the $2m$ -form, which is a gauge anomaly in $2m$ -dimensions, from the chiral anomaly Ch_m in $2m + 2$ dimensions is called the method of descent. It is said that the $2m + 2$ form Ch_m is related to the m -form P_{2m}^1 via the descent equations.

Example(Gauge anomaly and chiral anomaly in four dimensions)

Chiral anomaly in 4 dimensions is given by the 2nd Chern character:

$$\int d^4x \partial_\mu j_\chi^\mu = \int \frac{1}{16\pi^2} \text{Tr}[F \wedge F] \quad (50)$$

$$\Rightarrow j_\chi^\mu = \frac{1}{16\pi^2} \epsilon^{\mu\beta\gamma\delta} \text{Tr}[A_\beta \partial_\gamma A_\delta + \frac{2}{3} A_\beta A_\gamma A_\delta]. \quad (51)$$

Therefore, the chiral current is given by the Chern-Simons form.

To calculate the gauge anomaly in 4–dimensions, we will start with the 6–dimensional Chern-Simons form and calculate its BRST variation:

$$\text{Tr}[F \wedge F \wedge F] = d\text{Tr}[A \wedge dA \wedge dA + \frac{3}{2} A^3 dA + \frac{3}{5} A^5] \quad (52)$$

$$\delta_B Q_5 = \delta_B \text{Tr}[A \wedge dA \wedge dA + \frac{3}{2} A^3 dA + \frac{3}{5} A^5] \quad (53)$$

$$= \text{Tr}[dc[dAdA + \frac{1}{2} d(A^3)]] \quad (54)$$

$$= d\text{Tr}[c \wedge d(AdA + \frac{1}{2} A^3)] \quad (55)$$

$$= dP_4^1(c, A, F). \quad (56)$$

Therefore, the gauge current is given by:

$$\int d^4x \partial_\mu j_G^\mu = \int \text{Tr}[c \wedge d(AdA + \frac{1}{2} A^3)] \quad (57)$$

$$\partial_\mu j_G^\mu = \epsilon^{\alpha\beta\gamma\delta} \text{Tr}[\partial_\alpha (A_\beta \partial_\gamma A_\delta + \frac{1}{2} A_\beta A_\gamma A_\delta)]. \quad (58)$$

Diagrammatically, this term corresponds to triangle loop diagrams with external gauge fields, as was shown by Adler, Bell, and Jackiw. In general, the gauge anomaly in $2m$ –dimensions will correspond to $m + 1$ polygonal loop diagrams with external gauge fields.

The above procedure to calculate gauge anomalies works fine in flat space, where the (weak) Atiyah-Singer theorem relates the Chern character with the index of the Dirac operator (which is equal to the Euler characteristic of the space by the Poincaré-Hopf theorem). In generic manifolds, we will have to use a stronger version of the Atiyah-Singer theorem to calculate the chiral anomaly and then, by descent, the gauge anomaly. The Hirzebruch-Riemann-Roch theorem states that the Euler characteristic of a vector bundle E in a curved space $\mathcal{M}$ is given by:

$$\chi(\mathcal{M}, E) = \int_{\mathcal{M}} Td(T_{\mathcal{M}}) \wedge Ch(E), \quad (59)$$

where $T_{\mathcal{M}}$ is the tangent bundle of the manifold. The stronger version of the Atiyah-Singer index theorem further relates this Euler character to the index of the Dirac operator and hence to the chiral anomaly. In the next section, we will use this prescription to calculate Weyl anomaly on twistor space.

4.3 Weyl anomaly on twistor space

Remember that the Penrose transform relates positive helicity spin s fields with $\mathcal{O}(2s-2)$ line bundles, and negative helicity spin s fields with $\mathcal{O}(-2s-2)$ line bundles on twistor space. Therefore, the contribution of these fields to the anomaly polynomial is given by:

$$\int \frac{1}{2} Td(T_{\mathcal{PT}}) [Ch(\mathcal{O}(2s-2)) + Ch(\mathcal{O}(-2s-2))]_{(4,4)}. \quad (60)$$

Since twistor space is 3-complex dimensional, we are only interested in the $(4,4)$ form of the above equation to apply the method of descent. The invariant classes of the tangent bundle will be polynomials in $\partial V^\alpha \in \Omega^{(1,1)}(\mathcal{PT}, T_{\mathcal{PT}})$. We will express all the invariant classes in terms of the Chern class. The Todd class is given by [6]:

$$Td(T_{\mathcal{PT}}) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2 + \frac{1}{720}(-c_1^4 + 4c_2c_1^2 + 3c_2^2 + c_1c_3 - c_4) + \mathcal{O}((\partial V)^5). \quad (61)$$

Since the twistor tangent space $T_{\mathcal{PT}}^x$ is 3-dimensional, the fourth Chern class terminates $c_4 = 0$.

To calculate the Chern class of the line bundles $\mathcal{O}(2s-2)$ and $\mathcal{O}(-2s-2)$, we will make use of the canonical line bundle $\mathbb{K} \cong \mathcal{O}(-4)$. We can rewrite $\mathcal{O}(2s-2) = \mathbb{K}^{-(2s-2)/4}$ and $\mathcal{O}(-2s-2) = \mathbb{K}^{(2s+2)/4}$. The Chern class of the canonical line bundle satisfies the following property $c(\mathbb{K}^n) = 1 + nc_1(\mathbb{K})$. Furthermore, since the canonical bundle is the determinant bundle needed to define the volume form, we have $c_1(\mathbb{K}) = -c_1(T_{\mathcal{PT}})$. Therefore, we have the following simplification for the Chern classes: $c(\mathcal{O}(2s-2)) = 1 + \frac{(s-1)}{2}c_1$, $c(\mathcal{O}(-2s-2)) = 1 - \frac{(s+1)}{2}c_1$. We will now present the Chern characters in terms of the Chern classes using Newton's identities:

$$Ch(\mathcal{O}(2s-2)) = 1 + \frac{(s-1)}{2}c_1 + \frac{(s-1)^2}{2 \cdot 2^2}c_1^2 + \frac{(s-1)^3}{6 \cdot 2^3}c_1^3 + \frac{(s-1)^4}{24 \cdot 2^4}c_1^4 + \mathcal{O}(c_1^5) \quad (62)$$

$$Ch(\mathcal{O}(-2s-2)) = 1 - \frac{(s+1)}{2}c_1 + \frac{(s+1)^2}{2 \cdot 2^2}c_1^2 - \frac{(s+1)^3}{6 \cdot 2^3}c_1^3 + \frac{(s+1)^4}{24 \cdot 2^4}c_1^4 + \mathcal{O}(c_1^5) \quad (63)$$

Therefore, the anomaly polynomial is given by:

$$Td(T_{\mathcal{PT}})[Ch(\mathcal{O}(2s-2)) + Ch(\mathcal{O}(-2s-2))]_{(4,4)} = \frac{1}{5760}[(15s^4 - 30s^2 + 7)c_1^4 + 4(15s^2 - 7)c_1^2c_2 + 8c_1c_3 + 24c_2^2] \quad (64)$$

Note that the anomaly factors out in four separate terms c_1^4 , $c_1^2c_2$, c_1c_3 , and c_2^2 . We can call them the α , β , γ , and δ anomalies respectively. We have the following:

$$\alpha = (15s^4 - 30s^2 + 7) \quad (65)$$

$$\beta = 4(15s^2 - 7) \quad (66)$$

$$\gamma = 8 \quad (67)$$

$$\delta = 24 \quad (68)$$

In 4-dimensions, the Weyl anomaly factors out in three terms, the a -anomaly, c -anomaly, and the vacuum energy Λ . The claim is that α , β , γ , and δ linearly depend on a , c , and Λ . Notably, the anomaly factors out into four distinct parts (rather than three) in twistor

space. Euclidean flat twistor space can be written as $\mathbb{PT} \cong \mathbb{R}^4 \times \mathbb{CP}^1$, essentially having a celestial sphere at each point of $4D$ space. The extra fourth anomaly is due to this celestial sphere, which appears as an internal gauge direction in the $4D$ theory upon compactification.

One way to find the linear dependence of the four anomalies in terms of a , c , and Λ is to use the already known values (see [7]) for different spin particles and find the linear coefficients. The other way is to understand the meaning of the terms c_1^4 , $c_1^2 c_2$, $c_1 c_3$, and c_2^2 in terms of the $4D$ physics. To do that, it will be helpful to look at self-dual Einstein gravity in twistor space.

5 Self-Dual Einstein gravity, Standard Model gauge anomalies, and anomaly to diffeomorphism

The Riemann tensor can be viewed as a 2-form in the following sense:

$$R^{ab} := R^{ab}{}_{\mu\nu} dx^\mu \wedge dx^\nu = d\omega^{ab} + \frac{1}{2} \omega^a{}_c \wedge \omega^{cb} \quad (69)$$

where ω^{ab} is the spin connection, and the anti-symmetric Latin indices denote the local Lorentz frame. We can further decompose the Riemann tensor into the self-dual and anti-self-dual bits:

$$R^{AA'BB'} = R^{AB} \epsilon^{A'B'} + R^{A'B'} \epsilon^{AB}. \quad (70)$$

As is standard, we will use the following 2-forms constructed out of the tetrads as the basis for the ASD and SD 2-forms:

$$\Sigma^{AB} = e^A{}_{A'} \wedge e^{BA'}; \quad \Sigma^{A'B'} = e^{A'}{}_A \wedge e^{B'A}. \quad (71)$$

The (anti)self-dual Riemann tensor now takes the following form:

$$R^{AB} = C^{ABCD} \Sigma_{CD} + \Phi^{AA'BB'} \Sigma_{A'B'} + \frac{1}{12} R \Sigma^{AB} \quad (72)$$

$$R^{A'B'} = C^{A'B'C'D'} \Sigma_{C'D'} + \Phi^{AA'BB'} \Sigma_{AB} + \frac{1}{12} R \Sigma^{A'B'}. \quad (73)$$

Where, C^{ABCD} and $C^{A'B'C'D'}$ are the self-dual and anti-self-dual Weyl tensor respectively. $\Phi_{AA'BB'}$ is the Ricci tensor and R is the Ricci scalar. Based on this decomposition we can defined the following instanton backgrounds:

- Self-dual Weyl manifolds: This is given by $C^{A'B'C'D'} = 0$. We have already seen that the self-dual Weyl gravity action is given by a BF theory with F being the anti-self-dual Weyl tensor.
- Vacuum manifolds: These are manifolds with a vanishing Ricci tensor $\Phi_{AA'BB'} = R = 0$.

- Self-dual vacuum manifolds: These are manifolds with a vanishing anti-self-dual Weyl tensor and a vanishing Ricci tensor. Consequently, in these manifolds the anti-self-dual Riemann tensor vanishes $R^{A'B'} = 0$. We will now study the gravity action for such a theory.

The action for the theory describing a self-dual-vacuum manifold is given by the Plébanski action with a zero-cosmological constant:

$$S = \int_{\mathcal{M}} B_{A'B'} R^{A'B'} \quad (74)$$

$r^{A'B'}$ being zero implies that such a manifold has $SU(2)$ holonomy and is therefore a hyperkähler manifold. Such manifolds come with a quaternionic structure and consequently three symplectic forms. Without any loss of generality, we can choose a gauge such that the ASD spin connection vanishes $\omega^{A'B'} = 0$. The torsionless condition implies that $d\Sigma^{A'B'} = 2\omega^{(A'}_{C'} \wedge \Sigma^{C'B')}$, and therefore $d\Sigma^{A'B'} = 0$ modulo gauge. Therefore, the three symplectic two-forms are given precisely by Σ^{11} , Σ^{22} , and Σ^{12} (or linear combinations thereof).

5.1 SD Einstein gravity on twistor space

The action for SD Einstein gravity with a zero cosmological constant on twistor space was first given in [8]. We have seen already in section 2.2 that the infinity twistor for spacetimes with a cosmological constant is given by:

$$I_{\alpha\beta} = \frac{1}{2} \begin{bmatrix} \Lambda \epsilon^{A'B'} & 0 \\ 0 & \epsilon^{AB} \end{bmatrix}; \quad I^{\alpha\beta} = \frac{1}{2} \begin{bmatrix} \epsilon^{A'B'} & 0 \\ 0 & \Lambda \epsilon_{AB} \end{bmatrix}. \quad (75)$$

For asymptotically flat spaces we have:

$$I^{\alpha\beta} = \frac{1}{2} \begin{bmatrix} \epsilon^{A'B'} & 0 \\ 0 & 0 \end{bmatrix}. \quad (76)$$

Therefore, we can define a Poisson bracket on twistor space using this infinity twistor:

$$\{.,.\} = I^{\alpha\beta} \frac{\partial}{\partial Z^\alpha} \frac{\partial}{\partial Z^\beta} = \epsilon^{A'B'} \frac{\partial}{\partial \mu^{A'}} \frac{\partial}{\partial \mu^{B'}}. \quad (77)$$

Notice that the Poisson bracket is restricted purely along the fibres of twistor space $\mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1$ and has weight $\mathcal{O}(-2)$. In SD Einstein gravity with a zero cosmological constant, we will only study diffeomorphisms of twistor space that preserve this Poisson structure, such a theory is referred to as Poisson BF theory.

Just like before, the diffeomorphisms are given by deformations in the complex structure:

$$\bar{\nabla} = \bar{\partial} + \mathcal{L}_V, \quad (78)$$

where $V \in \Omega^{(0,1)}(\mathcal{PT}, T_{\mathcal{PT}})$, the specific form of V for Poisson BF theory is given by:

$$V = \epsilon^{A'B'} \frac{\partial h}{\partial \mu^{A'}} \frac{\partial}{\partial \mu^{B'}} = \{h,.\}, \quad (79)$$

where $h \in \Omega^{(0,1)}(\mathcal{PT}, \mathcal{O}(2))$, the weight $\mathcal{O}(2)$ has been taken to remove the weight $\mathcal{O}(-2)$ coming from the Poisson bracket. The corresponding Nijenhuis is given by:

$$N = \bar{\partial}V + \frac{1}{2}[V, V] \quad (80)$$

$$= \{\bar{\partial}h + \frac{1}{2}\{h, h\}, \cdot\}. \quad (81)$$

Where, $T := \bar{\partial}h + \frac{1}{2}\{h, h\} \in \Omega^{(0,2)}(\mathcal{PT}, \mathcal{O}(2))$. Note that $2 = 2(2) - 2$; therefore, h is a positive helicity graviton field. The action for self-dual Einstein gravity is given by:

$$S = \int_{\mathcal{PT}} g \wedge (\bar{\partial}h + \frac{1}{2}\{h, h\}), \quad (82)$$

where $g \in \Omega^{(3,1)}(\mathcal{PT}, \mathcal{O}(-2)) \cong \Omega^{(0,1)}(\mathcal{PT}, \mathcal{O}(-6))$ is a negative helicity graviton field, since $-6 = -2(2) - 2$. The equations of motion are given by:

$$\bar{\partial}h + \frac{1}{2}\{h, h\} = 0 \quad (83)$$

$$\bar{\nabla}g = \bar{\partial}g + \{h, g\} = 0. \quad (84)$$

The first equation guarantees the vanishing of the Nijenhuis tensor, thus making the complex deformation integrable. The second equation of motion is a background anti-self-dual field propagating in space.

Just like self-dual conformal gravity, this theory is invariant under the following gauge transformations:

$$\delta h = \bar{\partial}\chi + \{h, \chi\} \quad (85)$$

$$\delta g = \{h, \chi\} + \bar{\partial}\phi + \{h, \phi\}. \quad (86)$$

It is possible to gauge fix this action with the standard BRST procedure. The anomaly in this theory will be given by the BRST variation of the effective action.

5.2 Anomaly in Poisson BF theory and gauge theory

We use the index theorem again to compute the anomaly. The anomaly polynomial will be the $(4, 4)$ form of the given integral:

$$\int \frac{1}{2}Td(T_{\mathcal{PT}})[Ch(\mathcal{O}(2s-2)) + Ch(\mathcal{O}(-2s-2))](4,4). \quad (87)$$

For self-dual gauge theory with Lie algebra $\mathfrak{g}$ and Poisson BF theory this becomes:

$$\int \frac{1}{2}Td(T_{\mathcal{PT}}) \wedge Ch(\mathfrak{g}) \wedge [1 + Ch(\mathcal{O}(-4))](4,4) + \int \frac{1}{2}Td(T_{\mathcal{PT}}) \wedge [Ch(\mathcal{O}(-2)) + Ch(\mathcal{O}(-6))](4,4). \quad (88)$$

Let us simplify this expression. For the Chern character of the Lie algebra, we will demand that the gauge field lives in the adjoint representation, where only the even Chern characters are non-zero:

$$Ch(\mathfrak{g}) = \dim(\mathfrak{g}) + Ch_2(\mathfrak{g}) + Ch_4(\mathfrak{g}) + \dots \quad (89)$$

To calculate the Chern character of the line bundles we will use the fibration $\mathcal{O}(n) \rightarrow \mathcal{PT} \rightarrow \mathbb{CP}^1$. Note that we could not use this in the case of conformal gravity since the $\mathbb{CP}^1$ coordinates were getting mixed with the coordinates on the fiber. In the case of Einstein gravity, however, the preserved infinity twistor ensures that there is no such mixing. The Chern character of a line bundle over $\mathbb{CP}^1$ is given by:

$$Ch(\mathcal{O}(n)) = 1 + c_1(\mathcal{O}(n)) = 1 - n/2x, \quad (90)$$

where, x is the Fubini-Study metric on $\mathbb{CP}^1$, $x = \frac{dz \wedge d\bar{z}}{(1+|z|^2)^2}$. Since this is the top-form on $\mathbb{CP}^1$, $x^2 = 0$.

Finally, we need to calculate the Todd class of the tangent bundle. Let us express the Todd class in terms of the Chern class (just like we did section-4.3):

$$Td(T_{\mathcal{PT}}) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2 + \frac{1}{720}(-c_1^4 + 4c_2c_1^2 + 3c_2^2 + c_1c_3 - c_4) + \dots \quad (91)$$

The first Chern class of the canonical bundle (holomorphic top-form) $\mathbb{K} \cong \mathcal{O}(-4)$ is given by $c_1(\mathbb{K}) = 2x$. Since $\mathbb{K}$ is the determinant bundle of the tangent bundle $c_1(T_{\mathcal{PT}}) := c_1 = -c_1(\mathbb{K}) = 2x$. Note that $c_1^2 = 0$, therefore the Todd class simplifies to:

$$Td(T_{\mathcal{PT}}) = 1 + \frac{1}{2}c_1 + \frac{1}{12}c_2 + \frac{1}{24}c_1c_2 + \frac{1}{720}(3c_2^2 + c_1c_3) + \dots \quad (92)$$

where we have removed the polynomials in c_1 and also removed c_4 since $T_{\mathcal{PT}}$ is only three dimensional. Once again, we will use the fact that the $\mathbb{CP}^1$ coordinates do not mix with the fiber coordinates to simplify c_3 . Breaking the tangent bundle into the base tangent bundle $T_{\mathbb{CP}^1}$ and the normal part $T_{\mathcal{N}}$, we have:

$$\begin{aligned} c(T_{\mathbb{CP}^1} \wedge T_{\mathcal{N}}) &= c(T_{\mathbb{CP}^1}) \wedge c(T_{\mathcal{N}}) \\ &= c_3(T_{\mathcal{N}}) + c_2(T_{\mathcal{N}})c_1(\mathbb{CP}^1) + c_3(T_{\mathbb{CP}^1}) + c_1(T_{\mathcal{N}})c_2(\mathbb{CP}^1) \\ &= c_2(T_{\mathcal{N}})x \end{aligned} \quad (93)$$

TO get the final expression, we use the fact that $c_3(T_{\mathcal{N}})$ is zero since $\dim(T_{\mathcal{N}}) = 2$ and by the same logic $c_2(T_{\mathbb{CP}^1}) = c_3(T_{\mathbb{CP}^1}) = 0$. Therefore, the term c_1c_3 in td_4 get simplified to $c_1^2c_2(T_{\mathcal{N}}) = 0$. The Todd class is now given by:

$$\begin{aligned} Td(T_{\mathcal{PT}}) &= 1 + \frac{1}{2}c_1 + \frac{1}{12}c_2 + \frac{1}{24}c_1c_2 + \frac{1}{720}3c_2^2 + \dots \\ Td(T_{\mathcal{PT}}) &= 1 + \frac{1}{2}c_1 - \frac{1}{12}ch_2 - \frac{1}{24}c_1ch_2 + \frac{1}{20}ch_4 + \dots \end{aligned} \quad (94)$$

In the last line, we have replaced the Chern classes by the Chern character. Finally, bringing together all the pieces we have:

$$\begin{aligned} (1 + ch(\mathcal{O}(-4))) &= 1 + 1 - (-4)/2x = 2 + 2x = 2 - c_1 \\ (ch(\mathcal{O}(2) + ch(\mathcal{O}(-6)))) &= 1 - 2/2x + 1 - (-6)/2x = 2 + 2x = 2 - 2c_1 \\ \frac{1}{2}Td \wedge (ch(\mathcal{O}(2) + ch(\mathcal{O}(-6))))_{4,4} &= \frac{1}{20}ch_4 \\ \frac{1}{2}Td \wedge ch(\mathfrak{g}) \wedge (1 + ch(\mathcal{O}(-4)))_{4,4} &= \frac{1}{20}ch_4(\mathfrak{g}) + \frac{\dim(\mathfrak{g})}{20}ch_4 - \frac{1}{12}ch_2(\mathfrak{g})ch_2 \end{aligned} \quad (95)$$

The anomaly polynomial now becomes:

$$\mathcal{A} = \frac{\dim(\mathfrak{g}) + 1}{20} \int ch_4(T_{\mathcal{PT}}) + \frac{1}{20} \int ch_4(\mathfrak{g}) - \frac{1}{12} \int ch_2(\mathfrak{g}) ch_2(T_{\mathcal{PT}}) \quad (96)$$

The first term corresponds to the purely gravitational anomaly, the second term is the purely gauge anomaly, whereas the last term is the mixed gauge-gravity anomaly. Let us look at what these terms look like after applying the method of descent.

5.2.1 Purely gauge anomaly

Let's apply the method of descent to the gauge anomaly:

$$\begin{aligned} \mathcal{A} &= \frac{1}{20} \int ch_4(T) \\ &= \frac{1}{20} \int tr \partial(a(\partial a)^3) \\ &\rightarrow \frac{1}{20} \int tr(a(\partial a)^3) \\ &\rightarrow \frac{1}{20} \int tr(c(\partial a)^3) \end{aligned} \quad (97)$$

This anomaly corresponds to the following box diagram in twistor space:

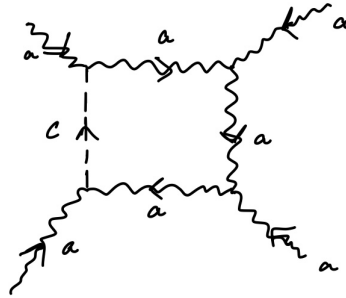


Figure 1: Purely gauge anomaly corresponds to calculating this diagram; c is the ghost field and a is the gauge field

5.2.2 Purely gravitational anomaly

The purely gravitational anomaly is given by:

$$\begin{aligned} \mathcal{A} &= \frac{\dim(\mathfrak{g}) + 1}{20} \int ch_4(\mathfrak{g}) \\ &= \frac{1}{20} \int tr \partial(s(\partial s)^3) \\ &\rightarrow \frac{1}{20} \int tr(s(\partial s)^3) \\ &\rightarrow \frac{1}{20} \int tr(\psi(\partial s)^3) \end{aligned} \quad (98)$$

Here $s^{A'}_{B'} = \partial^{A'} \partial_{B'} h$ is the connection on the tangent space and ψ is the corresponding ghost field. Notice that there is a factor of $\dim(g)$ in the numerator; this is because the gauge fields can also run in the loop with four external gravitons. The anomaly is given by the following diagrams:

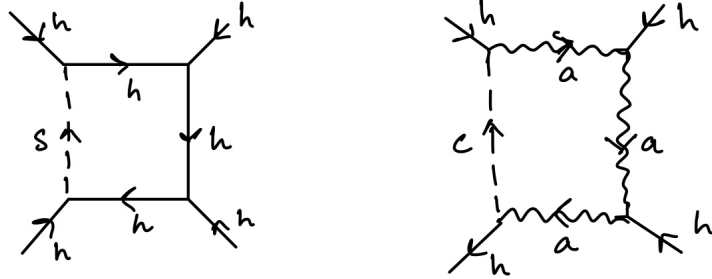


Figure 2: Purely gravitational anomaly corresponds to these two diagrams

5.2.3 Gauge-gravity mixed anomalies

The mixed anomaly comes from the following expression:

$$\mathcal{A} = -\frac{1}{12} \int ch_2(T) ch_2(\mathfrak{g}) \quad (99)$$

and it is depicted in figure(3).

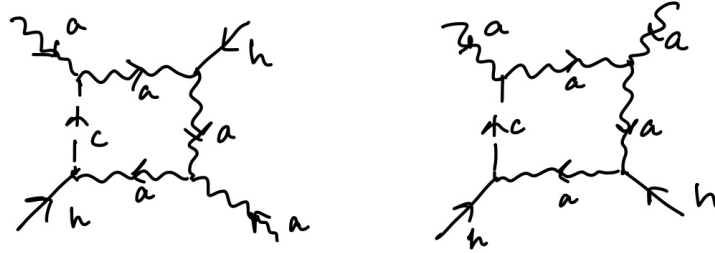


Figure 3: The mixed, gauge-gravitational anomaly

5.3 Getting spacetime anomalies

How to get spacetime anomalies from twistor space? Remember that in Euclidean signature, flat twistor space can be written as $\mathbb{PT} \cong \mathbb{R}^4 \otimes \mathbb{CP}^1$. If we compactify the extra $\mathbb{CP}^1$, we can get back to our 4D space. This corresponds to looking for terms in the anomaly polynomial that have a factor of x and removing it (essentially conformal compactification of the Riemann sphere). The remaining $(3, 3)$ form will correspond to the anomaly on 4D space. Pictorially this is described in figure(4).



Figure 4: Conformal compactification leads to the well-known triangle diagrams on space-time

6 Future work

We have discussed anomaly calculation for self-dual conformal spacetime and for self-dual Einstein spacetimes with a vanishing cosmological constant. It remains to calculate what these anomalies look like in self-dual Einstein spacetimes with a non-zero cosmological constant. Also, we have not discussed the method of anomaly cancellation. In [1, 4], the authors discuss how to cancel these anomalies using a scale field that has mass-dimension zero and a fourth-derivative kinetic term.

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